

Diverse Sources of C. difficile Infection Identified on Whole-Genome Sequencing

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BACKGROUND It has been thought that *Clostridium difficile* infection is transmitted predominantly within health care settings. However, endemic spread has hampered identification of precise sources of infection and the assessment of the efficacy of interventions. **METHODS** From September 2007 through March 2011, we performed whole-genome sequencing on isolates obtained from all symptomatic patients with *C. difficile* infection identified in health care settings or in the community in Oxfordshire, United Kingdom. We compared single-nucleotide variants (SNVs) between the isolates, using *C. difficile* evolution rates estimated on the basis of the first and last samples obtained from each of 145 patients, with 0 to 2 SNVs expected between transmitted isolates obtained less than 124 days apart, on the basis of a 95% prediction interval. We then identified plausible epidemiologic links among genetically related cases from data on hospital admissions and community location. **RESULTS** Of 1250 *C. difficile* cases that were evaluated, 1223 (98%) were successfully sequenced. In a comparison of 957 samples obtained from April 2008 through March 2011 with those obtained from September 2007 onward, a total of 333 isolates (35%) had no more than 2 SNVs from at least 1 earlier case, and 428 isolates (45%) had more than 10 SNVs from all previous cases. Reductions in incidence over time were similar in the two groups, a finding that suggests an effect of interventions targeting the transition from exposure to disease. Of the 333 patients with no more than 2 SNVs (consistent with transmission), 126 patients (38%) had close hospital contact with another patient, and 120 patients (36%) had no hospital or community contact with another patient. Distinct subtypes of infection continued to be identified throughout the study, which suggests a considerable reservoir of *C. difficile*. **CONCLUSIONS** Over a 3-year period, 45% of *C. difficile* cases in Oxfordshire were genetically distinct from all previous cases. Genetically diverse sources, in addition to symptomatic patients, play a major part in *C. difficile* transmission. (Funded by the U.K. Clinical Research Collaboration Translational Infection Research Initiative and others.)